Meal Planning and Grocery List Automation Web App

Christopher Perez

Advisor: Dr. Grabowski
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Abstract

This proposal outlines a web application that assists users with weekly meal planning and grocery preparation. The system will allow users to specify dietary preferences, the number of meals they want to plan, and other optional criteria such as cooking time or budget. Using this information, the application will generate a customized meal plan with recipes and produce a categorized grocery list based on the required ingredients. The goal of the project is to reduce the time and effort involved in deciding meals and organizing shopping. The proposal describes the planned features, development approach, and expected scope of the completed system, which will demonstrate skills in full-stack web development and practical application design

1 Introduction

For my senior capstone project, I plan to create a web application that helps people plan their meals and automatically build grocery lists. The app will let users enter their dietary preferences, how many meals they want for the week, and possibly extra features like their budget or how much time they have to cook. From there, it will generate a personalized meal plan with recipes and a categorized grocery list that makes shopping easier. The final product will be a working web app with a simple, user-friendly interface that I can demo and that real users can test.

I came up with this idea because, as a college student, I often struggle to balance time, money, and nutrition when it comes to eating. Planning meals ahead of time helps, but it takes a lot of effort to pick recipes, track ingredients, and make sure nothing goes to waste. I realized this isn't just my problem—many students and working adults deal with the same challenge. My goal is to build a system that automates this process so meal planning becomes less stressful and more efficient.

This project is also a great fit for my interests and career goals. I want to work in web development, and this idea lets me practice the entire development process from designing the interface and writing backend logic to integrating APIs and deploying the app. It's not just another website; it's a full-stack application that solves a real problem while letting me build something people might actually use. By the end of this project, I hope to have a functional application that demonstrates both my technical skills and my ability to design a practical solution to an everyday problem.

2 Background and Motivation

Meal planning and grocery organization have been popular topics for both consumers and developers leading to a lot of existing apps and services. Some well-known tools in this space include Mealime [1], Paprika [7], and Plan to Eat [8]. Each offers useful features but also has limitations that leave room for improvement.

Mealime focuses on quick and healthy recipes that generate grocery lists automatically, but its features are locked behind a subscription, and users have limited control over customizing meal plans beyond what the app suggests [1]. Paprika offers a recipe manager with grocery list generation but it is mainly a personal recipe storage tool and does not provide meal planning automation or dietary filtering [7]. Plan to Eat emphasizes meal calendar scheduling, but it requires users to manually add recipes, which can make initial setup time-consuming [8].

These existing solutions show that the idea of automating meal planning is useful, but many current options are either paid, not fully automated, or lack integration with grocery services. My proposed project aims to bridge these gaps by offering a free and personalized web app that automates meal planning based on user preferences, budget, and time constraints, while generating categorized grocery lists. In the future, I plan to explore API integration with grocery services like Walmart or Instacart to make the shopping process even more seamless.

My motivation for this project comes from personal experience. As a college student, I often find it difficult to eat well while managing time and expenses. I have tried apps like Mealime and Paprika but I found that they either didn't fit my routine or required more manual effort than I expected. That led me to realize that a helpful system should combine personalization, automation, and simplicity.

3 Preparation

I believe I am well prepared to take on this project based on both my coursework and my independent projects. The core Computer Science classes I've taken including CIS 201, CIS 203, and CIS 205 built the foundation for my programming and problem-solving skills. These courses covered fundamental concepts such as arrays, trees, loops, and conditional logic, which are essential for building backend algorithms. Most importantly, all of these classes placed a strong emphasis on code documentation and readability in which I've come to appreciate as a key part of professional software development. This focus on writing clear maintainable code will be especially valuable when building and testing a full-stack web application like my project.

Beyond coursework, I've also gained hands on experience through independent projects. One of my most relevant projects is a Personal Task Manager Web App that I designed and developed using React.js[2], Node.js[6]/Express[5], and MongoDB[3], with deployment on Heroku[10] and Netlify[4]. In that project, I implemented RESTful APIs for handling user data, created a persistent database, and configured a live production environment. This experience taught me how to structure a full-stack application, debug integration issues between the frontend and backend, and manage deployment all of which are directly applicable to my meal-planning web app.

4 Practice

This capstone brings together the core skills I've developed in classes like CIS 201, CIS 203, and CIS 205, such as programming, data structures, and documentation, and applies them in a larger real-world project. Unlike class assignments, this project will require me to design and manage an entire software system from start to finish including frontend, backend, and deployment.

It also goes beyond technical skills by helping me build professional abilities like project management, time organization, and clear communication through documentation.

5 Work and Work Products /OR/ Deliverables

The main deliverable for this project will be a functional web application that automates weekly meal planning and grocery list creation. The core features will include user profile and preferences, allowing users to specify dietary restrictions, preferred number of meals, and optionally cooking time or budget limits. The app will generate a weekly meal plan with recipes based on user input, and compile the required ingredients into a categorized grocery list, such as produce, dairy, and pantry items. Users will also be able to view recipe details including instructions, servings, and nutritional information. Additional features, such as budget constraints and time-based recipe filtering, may be added in future versions.

The system architecture will consist of a React.js[2] frontend for the user interface, a Node.js[6]/Express[5] backend to process requests and manage business logic, and a MongoDB[3] or PostgreSQL[9] database to store user data, recipes, and preferences. The backend will interact with a third-party recipe API (e.g., Spoonacular[11]) to retrieve meal and ingredient information. Communication between frontend and backend will occur via RESTful API calls, forming a clear workflow from user input to meal plan generation to grocery list output.

The completed project will include technical and user documentation, testing results, and deployment on a cloud platform such as Heroku[10] or Vercel[12], providing a live demo environment. The

final product will demonstrate how a full-stack web application can simplify meal planning and grocery management.

6 Administration

This senior project will be completed during the Spring 2026 semester. I plan to register for the Capstone Project course as part of my regular semester load of approximately 15 total credits. Project work will begin at the start of the semester in January 2026 and continue through the end of the term, with regular meetings held with my faculty advisor throughout that period.

The final project presentation will take place near the end of the semester, followed by submission of the final written report and documentation. All development, testing, and deployment activities will occur during this timeframe.

A Grading rubric (Senior Projects ONLY)

Grade	Description
4.0	Fully implemented and exceeds initial goals. Includes a functional React frontend, Node.js/Express backend, and MongoDB/PostgreSQL database. Users can create meal plans, view recipes, and generate categorized grocery lists. API integration works reliably, and the app is deployed online with full documentation.
3.5	All core features functional with only minor issues. Deployed and mostly stable. Optional features such as filtering or user accounts may be partially complete.
3.0	Meets main functional requirements. Users can generate meal plans and grocery lists, though the interface or deployment may have small issues. Documentation present but basic.
2.5	Some major features missing or incomplete, such as API integration or data storage. Deployment may not function. Limited documentation.
2.0	Minimal implementation demonstrating understanding of core concepts but lacking functionality, integration, and usability.

Table 1: Grading Rubric for Capstone Project

B Tentative schedule (Senior Projects ONLY)

References

[1] Mealime Inc. Mealime: Meal planner and recipes. https://www.mealime.com, 2025. Accessed on 2025-11-01.

[2] Meta Platforms, Inc. React.js: A javascript library for building user interfaces. https://react.dev/. Accessed on 2025-11-01.

[3] MongoDB Inc. Mongodb nosql database. https://www.mongodb.com/. Accessed on 2025-11-01.

[4] Netlify, Inc. Netlify web deployment platform. https://www.netlify.com/. Accessed on 2025-11-01.

[5] OpenJS Foundation. Express: Fast, unopinionated, minimalist web framework for node.js. https://expressjs.com/. Accessed on 2025-11-01.

[6] OpenJS Foundation. Node.js javascript runtime. https://nodejs.org/. Accessed on 2025-11-01.

[7] Paprika App. Paprika recipe manager. https://www.paprikaapp.com, 2025. Accessed: 2025-11-01.

[8] Plan to Eat. Meal planning done right. https://www.plantoeat.com, 2025. Accessed: 2025-11-01.

Week	Dates (Approx.)	Task / Milestone	Notes
1	Jan 13 – Jan 19	Meet with faculty advisor to confirm project scope and timeline	Establish communication schedule
2	Jan 20 – Jan 26	Finalize project requirements and design plan	Define key features and data flow
3–4	Jan 27 – Feb 9	Build backend structure (Node.js/Express, database schema)	Begin coding core logic
5–6	Feb 10 – Feb 23	Develop frontend framework (React setup, basic UI)	Connect frontend to backend
7	Feb 24 – Mar 1	Implement recipe and grocery list generation features	Integrate database and user input
8	Mar 2 – Mar 8	Midterm check-in with advisor	Review progress and adjust timeline
9–10	Mar 9 – Mar 22	Add user preferences, filtering, and account features	Expand backend endpoints
11	Mar 23 – Mar 29	Integrate third-party recipe or grocery API	Test and debug connections
12	Mar 30 – Apr 5	User testing and gather feedback	Improve usability and fix bugs
13	Apr 6 – Apr 12	Prepare final demo and documentation; **Final presentation and report submission**	Draft final report; Presentation date TBD

Table 2: Project Timeline (13 Weeks)

[9] PostgreSQL Global Development Group. Postgresql: The world's most advanced open source relational database. https://www.postgresql.org/, 2025. Accessed on 2025-11-01.

[10] Salesforce, Inc. Heroku cloud application platform. https://www.heroku.com/. Accessed on 2025-11-01.

[11] Spoonacular, Inc. Spoonacular recipe and food api. https://spoonacular.com/food-api, 2025. Accessed on 2025-11-01.

[12] Vercel, I. Vercel: Frontend deployment and hosting platform. https://vercel.com/, 2025. Accessed on 2025-11-01.